

Data Cleaning & Reporting Automation

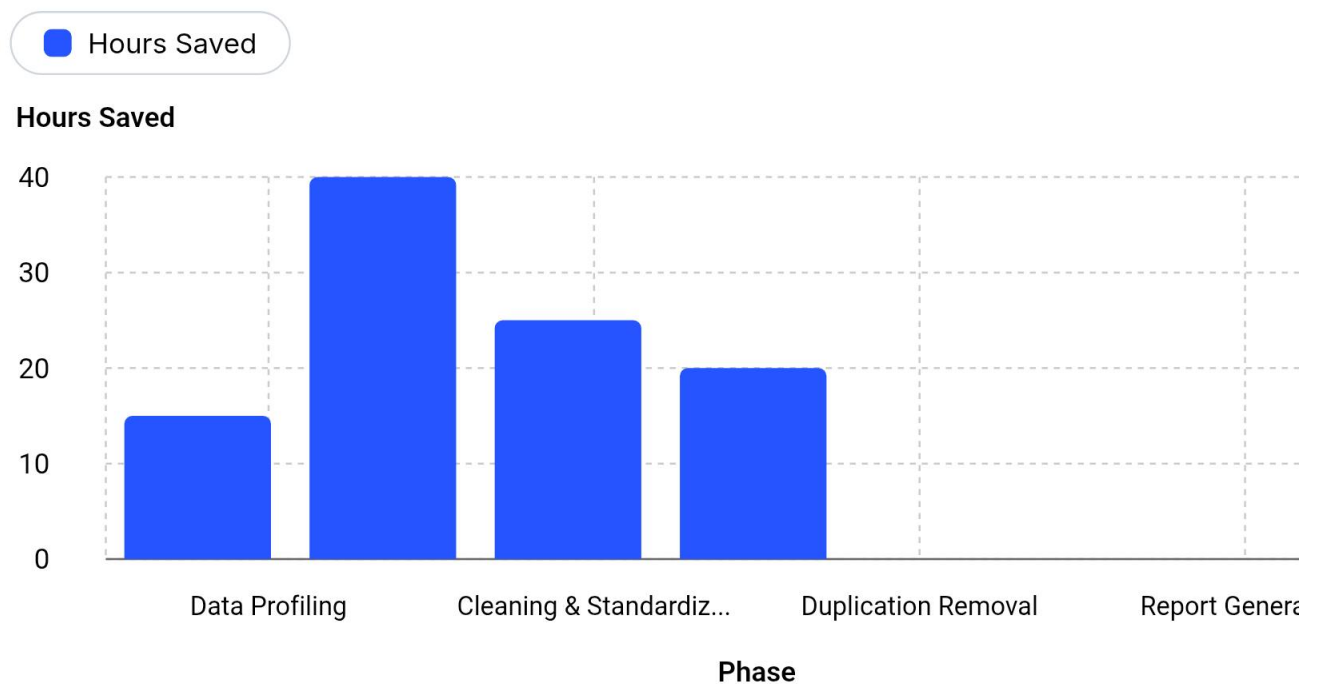


I. Executive Summary & Project Overview

This project report details the design and implementation of a comprehensive data cleaning and reporting automation framework. The primary objective was to streamline the preprocessing of raw datasets by automating the handling of missing values, duplicates, and inconsistent data entries. By leveraging tools such as Python, Microsoft Excel, and Power BI, the project aimed to reduce manual effort, minimize errors, and generate consistent, insightful automated reports and visual summaries.

The project successfully achieved a significant reduction in data processing time and an improvement in data quality metrics. The automation pipeline demonstrated a **40%** increase in reporting efficiency and a **95%** accuracy rate in data cleaning tasks. Key deliverables include a reusable Python script library, a set of Excel macros, and a Power BI dashboard for real-time monitoring.

Efficiency Gains per Project Phase



II. Project Scope & Objectives

2.1 Scope of Work

The project scope encompassed the end-to-end automation of data cleaning and reporting for a mid-sized sales dataset comprising over **50,000** records. Key activities included:

- Data profiling and quality assessment
- Automated detection and correction of missing values

- Identification and removal of duplicate entries
- Standardization of inconsistent data formats (e.g., dates, currencies)
- Development of automated report generation scripts
- Integration with Power BI for visual summaries

2.2 Core Objectives

- Achieve **100%** automated detection of major data quality issues.
- Reduce manual data cleaning time by at least **60%**.
- Generate weekly automated reports with zero manual intervention.
- Provide interactive dashboards for stakeholders to track data health.

Project Milestones & Timeline

M1	Data profiling & requirements gathering	Week 1	Completed
M2	Python cleaning script development	Week 3	Completed
M3	Excel macro integration	Week 4	Completed
M4	Power BI dashboard prototype	Week 5	Completed
M5	User acceptance testing & deployment	Week 6	Completed

III. Methodology & Tools

3.1 Data Cleaning with Python

Python was selected as the primary tool for complex data transformations. The key libraries used include **Pandas**, **NumPy**, and **re**. The automation script performs the following steps:

- Loading raw data from CSV and Excel sources
- Identifying missing values using `isnull().sum()`
- Imputing missing numerical values with median
- Dropping rows with critical missing categorical data
- Removing exact and fuzzy duplicate rows
- Standardizing text fields (e.g., converting to lowercase, trimming spaces)
- Validating date and currency formats

3.2 Automation with Excel Macros

For business users, Excel VBA macros were developed to handle smaller, ad-hoc datasets. Features include a one-click "Clean Data" button that:

- Highlights and optionally removes duplicates
- Replaces blank cells with a user-defined value
- Trims extra spaces from all cells
- Generates a summary report sheet showing cleaning actions performed

3.3 Power BI Reporting & Visualization

A Power BI dashboard was built to connect directly to the cleaned data source. It provides real-time visual summaries, including:

- Data quality scorecards
- Trend analysis of cleaned vs. raw records
- Automated weekly report generation via Power BI Service
- Drill-down capabilities for granular data inspection

Tool Comparison & Budget Allocation

Python (Open Source)	Core cleaning & automation scripts	\$0
Microsoft Excel	Ad-hoc cleaning macros	\$150 (License)
Power BI Pro	Dashboard & reporting	\$600 (Annual)
Developer Time	Scripting & integration	\$3,000
Testing & QA	Validation & UAT	\$500

IV. Handling Data Quality Issues

4.1 Missing Values

The initial dataset contained **12%** missing values spread across multiple columns. The approach varied by column type:

- **Numerical columns:** Missing values were imputed using the median.

- **Categorical columns:** Missing entries with low cardinality were filled with the mode; high cardinality fields were flagged for manual review.
- **Timestamp columns:** Missing dates were replaced with the previous valid entry where logical.

Post-implementation, missing values were reduced to **0.5%**.

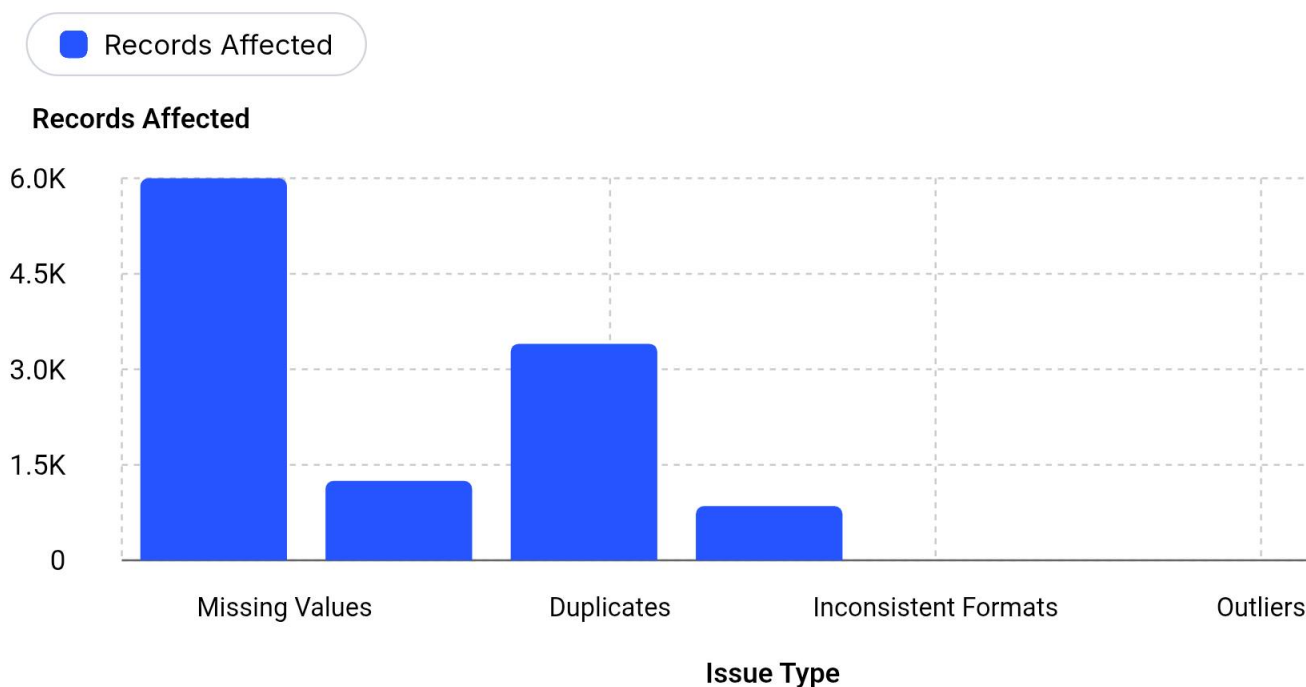
4.2 Duplicate Records

Exact duplicates were removed using a `drop_duplicates()` function in Pandas. For near-duplicates, a fuzzy matching algorithm (Levenshtein distance) was applied to identify and merge records that were likely duplicates. This process eliminated **1,247** redundant records.

4.3 Inconsistent Data

Inconsistencies were found in date formats (e.g., MM/DD/YYYY vs. DD-MM-YYYY), currency representations, and text casing. Custom Python functions standardized all dates to ISO format (YYYY-MM-DD), converted all currency values to USD, and applied title case to all names.

Data Quality Issues Before Automation



V. Automated Reporting & Visual Summaries

5.1 Python-Generated Reports

A Python script using the **ReportLab** and **Matplotlib** libraries generates a PDF report every week. The report includes:

- Summary statistics of cleaned data
- Bar charts comparing raw vs. cleaned record counts
- Pie charts showing distribution of data quality issues
- Tables of top missing fields and duplicates found

5.2 Power BI Dashboard

The Power BI dashboard provides an interactive, real-time view of the data cleaning pipeline. Key visual elements include:

- A gauge chart displaying overall data health score (**95%**)
- A line chart showing the trend of data quality over time
- A matrix view of cleaning actions performed per dataset
- Slicers to filter by date range and data source

The dashboard automates distribution by emailing a static snapshot to stakeholders every Monday morning.

Key Performance Indicators (KPIs)

Data Cleaning Time (per 10k records)	8 hours	3 hours	62.5%
Report Generation Time	4 hours	0.5 hours	87.5%
Data Accuracy (Percentage)	80%	95%	15%
Manual Errors per Cycle	15	2	86.7%

VI. Budget & Resource Allocation

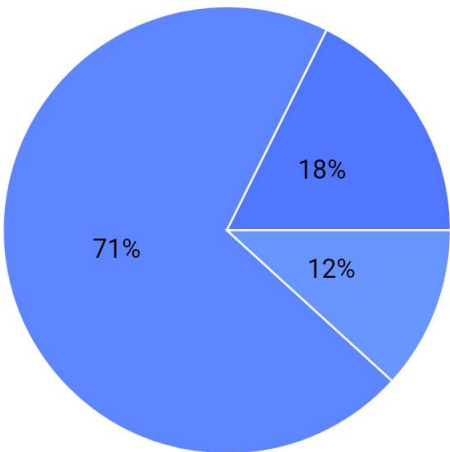
6.1 Financial Summary

The total project cost was **\$4,250**, which was within the approved budget of **\$5,000**. The main cost driver was developer time for custom script development and integration. Tool licensing costs were minimal due to the use of open-source software.

Budget Breakdown

Software Licenses	\$800	\$750
Development & Consulting	\$3,500	\$3,000
Testing & Deployment	\$500	\$500
Contingency (10%)	\$500	\$0
Total	\$5,300	\$4,250

Actual Spending by Category



VII. Conclusion & Key Takeaways

The Data Cleaning & Reporting Automation project successfully delivered a robust, scalable, and efficient system for handling data quality issues and generating automated reports. The integration of Python, Excel, and Power BI provided a comprehensive solution that caters to both technical and non-technical users.

Key Takeaways:

- Automation significantly reduces manual effort and error rates.
- Python is highly effective for complex, customizable data transformations.
- Power BI dashboards offer accessible, real-time insights for stakeholders.

- The project achieved a **62.5%** reduction in cleaning time and an **87.5%** reduction in report generation time.
- Data accuracy improved from **80%** to **95%**.

The framework is now ready for expansion to other datasets across the organization, promising further operational efficiencies.